

Goal. Figure out how to evaluate trig integrals

0. If we were using u -substitution and we set $u =$ one of the six trig functions, then $du =$?

$$u = \sin(x) \quad du =$$

$$u = \cos(x) \quad du =$$

$$u = \tan(x) \quad du =$$

$$u = \sec(x) \quad du =$$

$$u = \csc(x) \quad du =$$

$$u = \cot(x) \quad du =$$

1. Evaluate the following integrals using one of the substitutions from Exercise 0. You may need to use some trig identities and/or simplify first.

(a) $\int \tan(x) \sec^3(x) \, dx$

(b) $\int \tan^2(x) \sec^4(x) \, dx$

(c) $\int \sin(3x) \tan^2(3x) \, dx$

(d) $\int \cos^{5/2}(4-x) \sin^5(4-x) \, dx$

(e) $\int \cot^3(x) \csc(x) \, dx$

(f) $\int \frac{\cos^2(x)}{\sin^4(x)} \, dx$

2. Evaluate the integral by the following steps.

$$\int \frac{1}{x^4 \sqrt{x^2 - 4}} dx$$

(a) First, substitute $x = 2 \sec(\theta)$. Convert the integral completely into the variable θ . (Hint: If $x = 2 \sec(\theta)$ then what does dx equal?)

(b) Use the trig identity $\tan^2(\theta) + 1 = \sec^2(\theta)$ to simplify the denominator by making the part under the square root into a perfect square. (This kind of simplification is the goal of trig substitution.) Then evaluate the integral.

(c) Convert back to the variable x by using our trig substitution $\sec(\theta) = \frac{x}{2}$ and the definition of the trig functions in terms of a right triangle.

